AMAL ROY

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OBJECTIVE

Fourth-year Computer Science and Engineering student (CGPA: 9.03/10) with strong foundations in data structures, algorithms, and object-oriented design, backed by hands-on experience across full-stack development, systems programming, and applied machine learning. Seeking to contribute to a product engineering R&D team, building robust and scalable core software functionalities.

EDUCATION

BMS College of Engineering — B.E. in Computer Science and Engineering 2023 – Present
CGPA: 9.03/10
12th Grade, CBSE 94.6%
10th Grade, CBSE 92%

TECHNICAL SKILLS

Programming Languages: Python, C, C++, JavaScript, TypeScript, SQL (PostgreSQL), HTML/CSS
Core CS Fundamentals: Data Structures & Algorithms, Object-Oriented Programming & Design (OOAD), Database Management Systems (DBMS), Computer Networks, Network Security
Frameworks & Tools: React, Supabase, Tailwind CSS, OpenCV, WebAssembly (Emscripten), CMake, Git/GitHub, VS Code

PROJECTS

Federated Learning-Based Intrusion Detection System — *Semi-Supervised Learning, SDN/IoT Security*

- Built a Federated Learning-based Intrusion Detection System (IDS) for SDN-enabled IoT environments.
- Designed a semi-supervised anomaly detection framework using autoencoders to identify malicious traffic with limited labeled data.
- Implemented trust-based malicious client filtering and robust aggregation to defend the system against poisoning attacks.
- Optimized communication using cosine similarity for efficient client selection on non-IID IoT data.
- Evaluated the framework through literature-driven analysis, targeting a secure and scalable federated learning approach.

Real-Time Drowsiness Detection System — *OpenCV, dlib, SciPy*

- Developed a real-time drowsiness detection system processing live webcam video using OpenCV, dlib, and SciPy.
- Implemented 68-point facial landmark detection to accurately localize eye regions frame by frame.
- Computed the Eye Aspect Ratio (EAR) — a geometric measure based on distances between landmark points — to detect prolonged eye closure and signs of fatigue.
- Triggered visual alerts when EAR remained below a threshold across consecutive frames, reducing false positives from normal blinking.
- Built a lightweight, end-to-end real-time pipeline spanning video capture, face detection, feature extraction, and alerting.

Flashcard Generator — *React, Supabase, PostgreSQL, Tailwind CSS*

- Built a full-stack flashcard application with React on the frontend and Supabase (PostgreSQL) on the backend, using email authentication, protected routing, and row-level security enforced at the database level.
- Designed an AI-assisted flashcard-generation pipeline: extracted text from user-uploaded PDFs and images via PDF.js and Tesseract.js, then generated structured term-definition pairs using Groq's Llama 3.3 through Supabase Edge Functions.
- Implemented multiple-choice quizzes with per-user attempt history and scoring, alongside flashcard-set management and public read-only sharing for unauthenticated viewers.
- Added image uploads and print-friendly layouts; deployed on Vercel and Supabase with version-controlled SQL migrations for reproducible schema changes.

Fake News Detection — *BERT, LightGBM*

- Built a fake news classification system using BERT for contextual embeddings from news articles.
- Fine-tuned BERT to extract semantic representations, then used LightGBM as the final classifier for high accuracy and efficient inference.
- Designed a modular pipeline separating feature extraction and classification for easier maintenance and iteration.
- Evaluated performance using accuracy, precision, recall, and F1-score.

Beaglex — Agent-Based Market Simulator — *C++20, Genetic Algorithms, WebAssembly*

- Built a C++20 limit order book matching engine supporting limit, market and IOC orders with price-time priority.
- Simulated a market of 152 trading agents and validated it against statistical properties of real stock returns.
- Used a genetic algorithm to evolve trading strategies and studied how algorithmic trading affects market stability.
- Compiled the engine to WebAssembly for an interactive browser demo.

CERTIFICATIONS

Artificial Intelligence Foundation — Infosys Springboard | Introduction to Deep Learning — Infosys Springboard
Supply Chain Digitization — NPTEL | Research Methodology — NPTEL